A correction to Keil's algorithm for minimum convex decomposition of simple polygons without Steiner points

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Abstract

Keil's dynamic programming algorithm is the most efficient known exact algorithm for minimum convex decomposition of simple polygons without Steiner points. We show that when the input polygon contains collinear vertices, the algorithm may fail to find a feasible decomposition or return a non-minimum one. Two examples expose the underlying issue: certain faces of the canonical triangulation are not recognized as base triangles because some of their sides are not valid diagonals. To correct this issue, we extend the definition of a diagonal to include any segment joining two non-adjacent vertices that is contained in the closure of the polygon, allowing contact with its boundary. The revised formulation permits spiky subpolygons and degenerate base triangles, while leaving the remainder of the algorithm unchanged. We also revise the original Java demonstration program by modifying its visibility polygon routine to allow a viewpoint's line of sight to touch or overlap the boundary of the polygon.

Keywords: Simple polygons, Convex decomposition, Dynamic programming

1. Introduction

Let P be a simple polygon in the plane with n vertices p_0, p_1, ..., p_{n-1} in counterclockwise order, r of which are reflex vertices with interior angles greater than π. A minimum convex decomposition of P without Steiner points is a partition of P into the smallest possible number of convex regions, each of whose vertices is a vertex of P.

Relatively little research has addressed exact solution methods for this problem. Greene (1983) gave the first known exact algorithm with a running time of $O(n^2r^2)$. Keil (1985) subsequently proposed a dynamic programming algorithm that solves the problem optimally in $O(nr^2 \log n)$ time. Keil and Snoeyink (2002) later presented an efficient implementation that improves the running time to $O(nr^2)$. To the best of our knowledge, Keil's algorithm remains the most efficient known exact algorithm for this problem.

In the rest of the paper, we first briefly review Keil's dynamic programming algorithm. We then show that it may produce incorrect results when the input polygon contains collinear vertices. Finally, we provide a correction to both the algorithm and its implementation.

2. Brief review of Keil's algorithm

A *diagonal* d_{ik} of P is a segment $\overline{p_ip_k}$ joining two vertices p_i and p_k, with $k - i > 1$, whose relative interior lies in the interior of P. In other words, the endpoints of a diagonal are *strictly visible* to each other. It is straightforward to verify that every minimum convex decomposition without Steiner points uses only *valid diagonals*, each of which is incident to at least one reflex vertex.

Each valid diagonal d_{ik} defines a *subpolygon* P_{ik} of P with vertices p_i, p_{i+1}, ..., p_k in counterclockwise order. The *subproblem* associated with P_{ik} is to find a minimum convex decomposition of P_{ik} without Steiner points; its *weight* w_{ik} is the number of diagonals used in such a decomposition. The *size* of the subproblem is defined as the number of vertices of P_{ik}. For algorithmic convenience, we treat p_0 as a reflex vertex (if it is not reflex by definition) and the original edge $\overline{p_0p_{n-1}}$ as an artificial diagonal $d_{0(n-1)}$; hence, $P = P_{0(n-1)}$.

For a subproblem P_{ik}, there may be exponentially many convex decompositions attaining the weight w_{ik}. For a convex decomposition of P_{ik}, the convex region incident to d_{ik} is called its *base region*. Let a and b be the indices of the vertices adjacent to p_i and p_k, respectively, in the base region, where $a = b$ is possible. The *pair* (a, b) denotes an equivalence class of decompositions of P_{ik}. Keil's algorithm retains only the minimum convex decompositions with *narrowest pairs*, whose base regions do not properly contain the base region of any other minimum convex decomposition of P_{ik}.

For a subproblem P_{ik}, a triangle T_{ijk} with vertices p_i, p_j, and p_k is a *base triangle* if $i < j < k$ and each of its sides is either an edge of P or a valid diagonal. A minimum convex decomposition of P_{ik} obtained through a base triangle T_{ijk} uses

- a minimum convex decomposition of P_{ij} if $j - i > 1$,
- a minimum convex decomposition of P_{jk} if $k - j > 1$,
- diagonal d_{ij} if $j - i > 1$ and one of the following conditions holds: p_i is not reflex or T_{ijk} cannot be merged with the base region of any minimum convex decomposition of P_{ij}, and

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- diagonal d_{jk} if $k - j > 1$ and one of the following conditions holds: p_i is reflex or T_{ijk} cannot be merged with the base region of any minimum convex decomposition of P_{jk}.

The weight w_{ik} is obtained by minimizing, over all base triangles T_{ijk}, the sum of

- w_{ij} if $j - i > 1$,
- w_{jk} if $k - j > 1$,
- 1 if diagonal d_{ij} is used, and
- 1 if diagonal d_{jk} is used.

As noted above, among all minimum convex decompositions of P_{ik}, the algorithm retains only those with narrowest pairs.

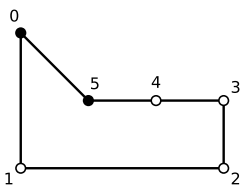
Along the recursive path for solving $P_{0(n-1)}$, the base triangles encountered form the *canonical triangulation* associated with the resulting decomposition. For a given convex decomposition of P that uses only valid diagonals, the canonical triangulation is uniquely defined as follows. For each convex region C in the decomposition, identify the reflex vertex of P with the lowest index among those contained in C, and connect it to every vertex of C with a higher index. If C is not yet triangulated, connect the vertex with the highest index of C to all remaining vertices of C.

The subproblems are solved in order of increasing size. Subproblems of equal size are processed in counterclockwise order, as are the base triangles of each subproblem. The subproblems and their base triangles can be enumerated in $O(nr^2)$ time, and the total number of stored narrowest pairs is also $O(nr^2)$. For merge tests between T_{ijk} and P_{ij}, the narrowest pairs of P_{ij} are scanned in counterclockwise order; for merge tests between T_{ijk} and P_{jk}, those of P_{jk} are scanned in clockwise order. Since each scan is monotone, each pair is passed at most once in each direction. Consequently, the overall running time is $O(nr^2)$.

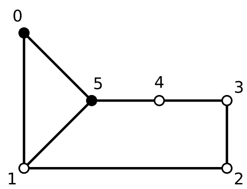
3. Examples and analysis

We now present two examples showing that, in the presence of collinear vertices, Keil's algorithm can fail to find a feasible decomposition or return a non-minimum one.

Fig. 1 shows a simple polygon with six vertices, where p_0 and p_5 are reflex vertices, and the valid diagonals are d_{25}, d_{15}, and d_{05}. When solving the first subproblem, P_{25}, the algorithm cannot find a base triangle because neither $\overline{p_2p_4}$ nor $\overline{p_3p_5}$ is a valid diagonal. Indeed, P_{25} is a quadrilateral with three collinear vertices. Although P_{25} is geometrically equivalent to a triangle and is therefore convex, the algorithm fails to recognize it as such.



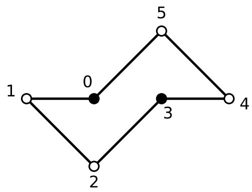
(a) Input polygon.



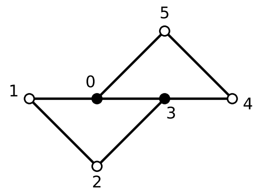
(b) Minimum convex decomposition.

Fig. 1. Example 1.

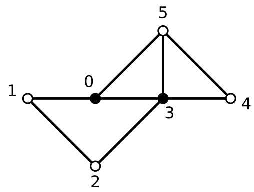
Fig. 2 shows another simple polygon with six vertices, where p_0 and p_3 are reflex vertices, and the minimum convex decomposition uses only the diagonal d_{03}. However, Keil's algorithm returns a convex decomposition using two diagonals, d_{03} and d_{35}. Table 1 details the algorithm's execution on this instance. It lists the convex decomposition computed for each subproblem, where | denotes separation by a diagonal. For the final subproblem P_{05}, the algorithm considers the only base triangle T_{035}. It includes d_{03} because T_{035} cannot be merged with the base region of any minimum convex decomposition of P_{03}, and it includes d_{35} because p_0 is reflex. It therefore returns a non-minimum convex decomposition with two diagonals.



(a) Input polygon.



(b) Minimum convex decomposition.



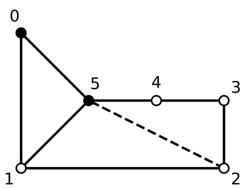
(c) Result of Keil's algorithm.

Fig. 2. Example 2.

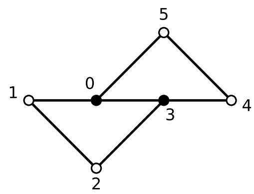
Table 1
Details of the algorithm's execution for Example 2.

Size	Subproblem	Decomposition	Pair	Weight
3	P_{02}	T_{012}	$(1,1)$	0
3	P_{35}	T_{345}	$(4,4)$	0
4	P_{03}	$P_{02}T_{023}$	$(1,2)$	0
6	P_{05}	$P_{03} \mid T_{035} \mid P_{35}$	$(3,3)$	2

Recall the construction of the canonical triangulation of a convex decomposition that uses only valid diagonals. Fig. 3 shows the canonical triangulations of the minimum convex decompositions for the two examples. Keil's algorithm fails on these instances because certain faces of the canonical triangulations—namely, T_{235} in Example 1 and T_{045} in Example 2—are not recognized as base triangles.



(a) Example 1.

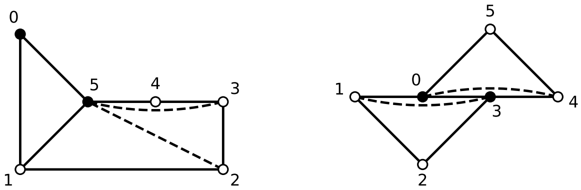


(b) Example 2.

Fig. 3. Canonical triangulations for the two examples.

4. Correction to the algorithm

To handle collinear vertices, we extend the set of diagonals to include segments that may touch or overlap the boundary of the polygon. This extension ensures that every face of the canonical triangulation contains exactly three vertices. Fig. 4 shows the resulting extended canonical triangulations. In Example 1, the additional diagonal d_{35} induces the degenerate triangle T_{345}. This enables the algorithm to reach P_{25} from P_{35} through T_{235} and thus find a minimum convex decomposition. In Example 2, the additional diagonals d_{13} and d_{04} induce the degenerate triangles T_{013} and T_{034}, respectively. The algorithm can then reach P_{04} from P_{03} through T_{034} and subsequently reach P_{05} from P_{04} through T_{045}, thereby obtaining a minimum convex decomposition.



(a) Example 1. (b) Example 2.

Fig. 4. Extended canonical triangulations for the two examples.

More precisely, we redefine a diagonal d_{ik} of P to be a segment $\overline{p_i p_k}$ with $k - i > 1$ that is contained in the closure of P, where contact with the boundary of P is allowed. Subpolygons and base triangles are defined with respect to this extended set of diagonals. Remark that this extension may introduce spiky subpolygons and degenerate base triangles. The remainder of the algorithm remains unchanged. Tables 2 and 3 detail the execution of the corrected algorithm on the two examples.

Table 2
Details of the corrected algorithm's execution for Example 1.

Size	Subproblem	Decomposition	Pair	Weight
3	P_{35}	T_{345}	$(4, 4)$	0
4	P_{25}	$T_{235}P_{35}$	$(3, 4)$	0
5	P_{15}	$T_{125}P_{25}$	$(2, 4)$	0
6	P_{05}	$T_{015} \mid P_{15}$	$(1, 1)$	1

Table 3
Details of the corrected algorithm's execution for Example 2.

Size	Subproblem	Decomposition	Pair	Weight
3	P_{02}	T_{012}	$(1, 1)$	0
3	P_{13}	T_{123}	$(2, 2)$	0
3	P_{35}	T_{345}	$(4, 4)$	0
4	P_{03}	$P_{02}T_{023}$	$(1, 2)$	0
5	P_{04}	$P_{03} \mid T_{034}$	$(3, 3)$	1
6	P_{05}	$P_{04}T_{045}$	$(3, 4)$	1

We also revise the original Java demonstration program of Keil and Snoeyink (2002). In particular, its *visibility polygon* routine now permits a viewpoint's line of sight to touch or overlap the boundary of the polygon. The corrected implementation is publicly available at https://github.com/jinboszu/mcd.

5. Conclusion

We have identified a degeneracy caused by collinear vertices that invalidates the original formulation of Keil's dynamic programming algorithm for minimum convex decomposition of simple polygons without Steiner points. Extending the definition of a diagonal restores the missing base triangles and thereby corrects the algorithm. The accompanying revision of the demonstration program makes the implementation consistent with the corrected formulation.

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